

1		1	
2		2	
3		3	
GND		GND	
5		5	
SCK		UART	
SDI		EN1	
SDO		END	
SCL		A1	
SDA	J1	A2	J2
DIR	FireBeetle 2	CS1	FireBeetle 2
STEP		CS2	
DIAG		CS3	
INDEX		CS4	
TX2		CS5	
RX2		CS6	
TX1			
RX1			

The diagram illustrates the pin connections for the J27 Expander L and J28 Expander R. The J27 Expander L has pins 3v3, GND, SCL, SDA, 5, MS2, MS1, 8, 9, 10, 11, 12, 13. The J28 Expander R has pins 1, 2, D0, D1, D2, 6, 7, 8, 9, 10, 11, 12, 13. The diagram shows the internal connections between these pins and the corresponding pins on the J27 and J28 headers.

Diagram showing the pin connections for the MAX31856 module. The connections are as follows:

- Module J8:** H1 (MAX31856) to VCC, H2 (MAX31856) to GND, H3 (MAX31856) to SCK, H4 (MAX31856) to SDO, H5 (MAX31856) to SDI, H6 (MAX31856) to CS1, H7 (MAX31856) to CS2, H8 (MAX31856) to CS3.
- Module J15:** H1 (MAX31856) to VCC, H2 (MAX31856) to GND, H3 (MAX31856) to SCK, H4 (MAX31856) to SDO, H5 (MAX31856) to SDI, H6 (MAX31856) to CS1, H7 (MAX31856) to CS2, H8 (MAX31856) to CS3.
- Module J16:** H1 (MAX31856) to VCC, H2 (MAX31856) to GND, H3 (MAX31856) to SCK, H4 (MAX31856) to SDO, H5 (MAX31856) to SDI, H6 (MAX31856) to CS1, H7 (MAX31856) to CS2, H8 (MAX31856) to CS3.
- Module J17:** H1 (MAX31856) to VCC, H2 (MAX31856) to GND, H3 (MAX31856) to SCK, H4 (MAX31856) to SDO, H5 (MAX31856) to SDI, H6 (MAX31856) to CS1, H7 (MAX31856) to CS2, H8 (MAX31856) to CS3.

3v3 1
GND 2
3
4 J19 RS232 LV
RX1 5
TX1 6

1
2
3 J20 RS232 HV
4
TXFC1 5
RXFC1 6

3v3 1
GND 2
3
4 J21 RS232 LV
RX2 5
TX2 6

1
2
3 J22 RS232 HV
4
TXFC2 5
RXFC2 6

Diagram showing the pin connections for H21, H22, H23, and H24. Each component has a red circle with a white dot representing a pin. H21 and H22 are connected to DE9 1 and DE9 4 respectively. H23 and H24 are connected to DE9 2 and DE9 5 respectively. The connections are shown as green lines with red dots at the component pins and red lines with red dots at the DE9 connector pins.

H21 DE9 1

H22 DE9 4

H23 DE9 2

H24 DE9 5

TXFC1 1

2

3

GND 4

5

J29

DE9 2

RXFC1 6

20V 7

GND 8

9

TXFC2 1

2

3

GND 4

5

J24

DE9 2

RXFC2 6

20V 7

GND 8

9

Diagram showing the pin connections for the I2C Sensor 1, I2C Sensor 2, I2C Sensor 3, and I2C Sensor 4. The connections are as follows:

- I2C Sensor 1:** GND 1, 3v3 2, SDA 3, SCL 4
- I2C Sensor 2:** GND 1, 3v3 2, SDA 3, SCL 4
- I2C Sensor 3:** GND 1, 3v3 2, SDA 3, SCL 4
- I2C Sensor 4:** GND 1, 3v3 2, SDA 3, SCL 4

The diagram also shows the connections for the I2C Sensor 1, I2C Sensor 2, I2C Sensor 3, and I2C Sensor 4. The connections are as follows:

- I2C Sensor 1:** GND 1, 3v3 2, SDA 3, SCL 4
- I2C Sensor 2:** GND 1, 3v3 2, SDA 3, SCL 4
- I2C Sensor 3:** GND 1, 3v3 2, SDA 3, SCL 4
- I2C Sensor 4:** GND 1, 3v3 2, SDA 3, SCL 4

 GND1 TestPoint GND	 3v3 TestPoint 3v3	 SCL1 TestPoint SCL
 5V1 TestPoint 5V	 20V1 TestPoint 20V	 TX1 TestPoint TX1
 SCK1 TestPoint SCK	 SDI1 TestPoint SDI	 RX1 TestPoint RX1
 SDO1 TestPoint SDO	 SDA1 TestPoint SDA	 TX2 TestPoint TX2
 Analog1 TestPoint A1	 Analog2 TestPoint A2	 RX2 TestPoint RX2

Diagram showing four mounting holes labeled H13, H14, H17, and H18, each connected to GND.

3v3	1	
GND	2	
DIR	3	
STEP	4	
MS1	5	
MS2	6	
DIAG	7	
INDEX	8	
UART	9	
EN1	10	

J18
 TMC2209

Id: 1/1